ENN 581 – Semester 2 – 2026
Week 8 – Dynamics

1. A manipulator with a single degree of freedom is defined as follows:

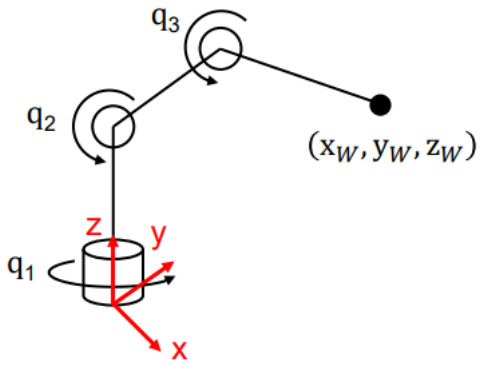
   It has a total mass m=1kg. The centre of mass is positioned at ${}^1P_c = \begin{bmatrix}3\\0\\0\end{bmatrix}$.

   The inertia tensor is given as ${}^C I = \begin{bmatrix}1 & 0 & 0\\0 & 2 & 0\\0 & 0 & 3\end{bmatrix}$.

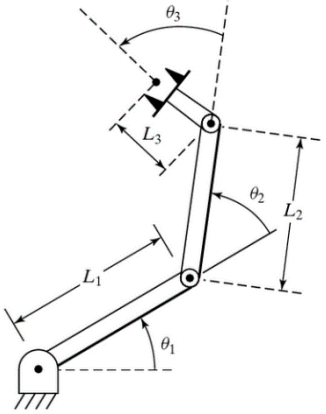
   The joint is initially at rest. It moves according to a function $q(t) = bt + ct^2$.

   Give expressions for the angular acceleration of the link and the linear acceleration of the centre of mass in terms of the link frame {1}.

2. Consider the 3D RRR manipulator shown below. Derive the dynamic equations of motion. Give expressions for the joint torques if a trajectory is prescribed.



3. Consider the 3-link planar manipulator shown below.



   The geometry of the robot is given as L1 = 3m, L2 = 2m, L3 = 1m. The robot starts from rest where the end effector pose is given as (5,1,90°). We would like to move it to (-3,3, 180°) and come to a rest there.

   a. Using the robotics toolbox, first describe the kinematics of the robot.
   b. Say, we want to use cubic polynomials to describe the trajectory above. Determine a suitable trajectory. Plot the joint positions, velocities and accelerations as a function of time.

c. Extend your analysis to now compute the joint torques at these instances. The masses are given as m1 = 3kg, m2 = 2kg, m3 = 1kg. Assume that they are concentrated at the distal ends of each link.